

Rocket Launch Dataset

*Looking Back to
Launch Forward:
Data to Predict the
Future of Spaceflight*

DSCI 511
Final Project

Innocent Gumunyu
Jillian Kunze
Ryan Peters
Phillip Roman

Image source:
<https://www.nasa.gov/humans-in-space/spaceships-and-rockets/commercial-crew-rockets>



Introduction and Purpose of Dataset



Why This Project?

Space launch data chronicles humanity's 68-year journey beyond Earth – from Sputnik and the Space Race through today's commercial revolution. This historical record has remained fragmented across sources; we set out to acquire, unify, and prepare it for analysis.



What We Acquired:

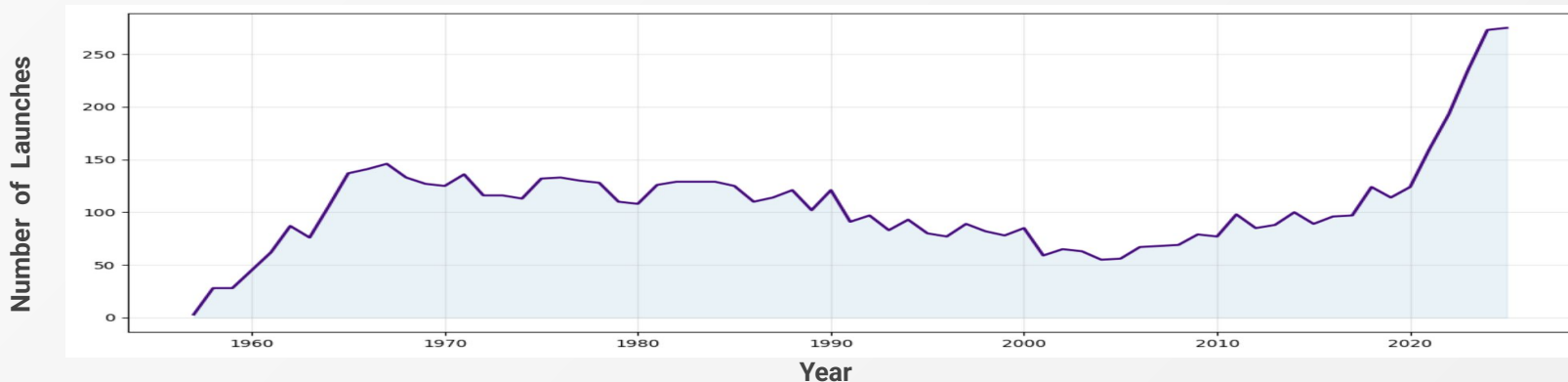
7,336 launches (1957-2025) | **341MB** raw data | **39 parameters** | Domains: **Rocket + Launch + Mission**



How We Built It:

API pagination with rate-limiting → Extraction from heavily nested JSON → Preprocessing and merging three parameter-specific dataframes

Historical Launch Coverage: Sputnik (1957) through present (Nov. 2025)



Potential Users and Applications

- Identify factors correlated with mission outcomes and predict space launch success
- Support predictive modeling, performance tracking, and comparative analysis
- Potential usage for academic and commercial audiences:
 -  **Academia and Research:** research historical trends, explore patterns in launch success/failure
 -  **Commercial Space Industry:** analyze competitors, develop business proposals, identify best practices
 -  **Regulatory and Government Agencies:** create data-driven guidelines, evaluate performance
 -  **Investors and Market Analysts:** assess current trends, compare launch providers, make investment decisions

Source of Data

Launch Library 2 API (The Space Devs)



<https://ll.thespacedevs.com/2.3.0/launches/>

- Free, public API (no authentication key required)
- 7,336+ historical launches (1957-2025)
- Detailed parameters: rockets, missions, locations, outcomes
- Well-documented RESTful endpoints
- Active maintenance and regular updates

Note: Wikipedia API was evaluated as a potential supplementary source for addressing data gaps but was not utilized in final dataset

Approach to Acquiring Data

STRATEGIES EVALUATED

- **Time-Filtering** (1-year increments)
Challenge: High-launch years (e.g., 2023-2024) exceeded 100-launch batch limits
- **Time-Based Pagination** (5-20 year spans)
Challenge: Inconsistent temporal distribution created uneven chunk sizes
- **Full Pagination** ✓ (selected approach)
Rationale: Reliable, complete coverage regardless of temporal distribution

PAGINATION WITH RATE-LIMIT HANDLING

- **14 requests/hour** (self-imposed limit, API allows 15/hour)
- **1 hour 30 seconds** buffer between batches
- **60-second timeout** threshold (vs. default 30 seconds)
- **Sequential "next" URL extraction** from API responses
- **100 launches per request** (API maximum)

Result: 74 batches → 7,336 launches collected over ~5 hours



Approach to Preprocessing Data

- Split into three categories: parameters related to the Rocket, Launch, and Mission
- Required searching through nested dictionary with many levels to find relevant parameters
- Steps in preprocessing each sub-dataset:
 - Read in raw JSON zipfile
 - Iterate over each launch to save parameters in table, and account for missing info
 - Inspect as Pandas dataframe
 - Save as TSV
- Then, merge on launch ID!
 - Two outer merges, require aligning ID column names

```
if launch['rocket']['configuration']['families']:
    manuf_name = launch['rocket']['configuration']['families'][0]['manufacturer'][0]['name']
    manuf_country = launch['rocket']['configuration']['families'][0]['manufacturer'][0]['country'][0]['name']
    manuf_type = launch['rocket']['configuration']['families'][0]['manufacturer'][0]['type']['name']
else:
    manuf_name = None
    manuf_country = None
    manuf_type = None
```

The Constructed Dataset

- Data for 7,336 launches with 39 columns
- Mix of text, numerical, and Boolean information
- Combination of Rocket, Launch, and Mission-related data provides many options for user to select based on their interests
- Nulls in 21 columns

File	Rows	Columns	File Size
Merged Dataset	7336	39	4.6 MB
Rocket Parameters	7336	15	1 MB
Launch Parameters	7336	16	2.1 MB
Mission Parameters	7336	10	2 MB

Rocket_name	Manufacturer_name	Manufacturer_country	Manufacturer_company_type	Reusability	Min_no_stages	Max_no_stages	Rocket_length	Rocket_diameter
Long March 7A	China Aerospace Science and Technology Corpora...	China	Government	False	NaN	NaN	NaN	NaN
Ariane 62	Arianespace	France	Commercial	False	2.0	2.0	63.0	5.40
Electron	None	None	None	False	2.0	3.0	18.0	1.20
Falcon 9 Block 5	SpaceX	United States of America	Commercial	True	1.0	2.0	70.0	3.65

Distribution Approach

- **Repository Access**

- Users can obtain the project by cloning our GitHub repository, which contains all scripts and resources.

- **Example Data**

- A folder of example data generated from our initial API calls is included in our repository.

- **Reproducible Data Retrieval Process**

- Users may run the provided notebooks to generate their own updated dataset at any time.
 - Documentation on how to run the notebooks provided in [README.md](#)
 - Notebook dependencies are provided in [requirements.txt](#) file
- Option to also use launchmodulev1 script that replicates notebook workflow, packaging API requests into simple, reusable functions for users who want to automate monthly data collection.

Discussion of Access Rights

- **Source API: Launch Library 2 (The Space Devs)**
 - Public API with no documented usage restrictions
 - Educational/research use permitted
 - No API key required for public endpoints
- **Our Contributions:**
 - Data collection methodology (pagination, rate limiting)
 - Cleaning and parameter extraction
 - Integration and documentation
- **Sharing Rights:**
 - ✓ Can share processed datasets
 - ✓ Can share collection/cleaning code
 - ✓ Must credit Launch Library 2 as data source
 - ✓ Launch IDs enable full reproducibility

Issues and Limitations

- **API Related Issues**

- No response from initial API provider, requiring alternative sources.
- Launch Library 2 API limited us to only 15 calls per hour with a maximum of 100 launches per call causing long collection times.
- Network timeouts during collection of Data due to connectivity issues.

- **Data Related Issues**

- Missing data from early launches
- Some string variables contain commas, which lead us to using TSV as our desired format to save the data.
- Not all variables selected relate to each rocket (eg. Payload Mass to varying orbits), led to combining variables to make new columns
- Time constraints preventing us from finding more variables.
- Use of Wikipedia API for missing values

- **Repository Issues**

- Github repository management with multiple users introduced merge conflicts

Team and Contributions

Team Member	Background	Key Contributions
Jillian	Physics; Science Communication	Original API requests & time filtering; Rocket/launch parameter cleaning
Ryan	Applied Mathematics; DS Certificate	Mapping nested level variables; GitHub management; Wikipedia webscraping;
Innocent	Statistics; DS Certificate	Hybrid pagination-time filtering; Launch parameter cleaning
Phillip	Aviation Mgmt; SATCOM Operator	Pagination calls; Mission parameter cleaning; DataFrame merging

References

1. The Space Devs. "The Space Devs." Accessed November 21, 2025. <https://thespacedevs.com/>.
2. The Space Devs. "LaunchLibrary API v2.3.0." Accessed November 21, 2025. <https://ll.thespacedevs.com/docs/>.
3. Next Spaceflight. "Launches." Accessed November 21, 2025. <https://nextspaceflight.com/launches/>.
4. The Space Devs. "Launch Library 2 - Frequently Asked Questions (FAQ)." *GitHub*. Accessed November 21, https://github.com/TheSpaceDevs/Tutorials/blob/main/faqs/faq_LL2.md.
5. The Space Devs. "Launch List" [Sample JSON Response]. Accessed October 12, 2025. <https://ll.thespacedevs.com/2.3.0/launches/?format=json>.

Questions?



Image source:
<https://www.nasa.gov/humans-in-space/spaceships-and-rockets/commercial-crew-rockets/>